PROJECT 1: Supply Chain Performance Analysis

Dataset information: The dataset includes fields such as:

- Approximately 180,519 records.
- More than 50 columns. • Order and shipping information. • Customer and product information.
- Delivery status.
- Shipping methods.
- Financial information.
- Regional information.

Column	Meaning
Order Id	Unique order identifier
Days for shipping (real)	Actual shipping duration
Days for shipment (scheduled)	Expected shipping duration
Late_delivery_risk	Whether delivery was late
Delivery Status	Delivery outcome
Shipping Mode	Shipping method
Order Region	Destination region
Sales	Order sales value
Benefit per order	Profit contribution

Questions

1 - Basic Problems

1. What is the total number of orders?
2. What is the total revenue?
3. What is the average actual shipping time?
4. What is the average scheduled shipping time?
5. What percentage of shipments were delivered late?
6. How many shipments were delivered on time?
7. What are the different shipping methods?
8. Which region has the highest number of orders?

Questions

2 - Intermediate Problems

9. Which shipping method has the highest late-delivery percentage?
10. Which regions experience the most delivery delays?
11. What is the average shipping time for each shipping method?
12. Which product categories experience the most delays?
13. Which month has the highest number of delayed deliveries?
14. Which customer segment generates the most revenue?
15. Is there a relationship between shipping duration and delivery delays?

3 - Advanced Problems

16. Identify the top 10 regions with the highest late-delivery rates.
17. Find the shipping methods whose actual delivery time exceeds their scheduled delivery time.
18. Calculate the average delivery-time difference by region.
19. Identify regions where revenue is high but delivery performance is poor.
20. Find the top five product categories with the highest delivery risk.
21. Analyse monthly delivery performance and identify seasonal patterns.
22. Identify the biggest operational contributors to late deliveries.
23. Compare shipping performance across different regions and shipping methods.
24. Identify whether certain shipping methods consistently underperform.
25. Recommend three operational improvements based on your findings.